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bapanapalli

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Editor

exp5.m exp5b.m exp6a.m exp6b.m exp7a.m

/MATLAB Drive/exp7a.m

```
clc;
clear;
close all;

Nsc = 24;          % Number of subcarriers
Nsym = 14;         % Number of OFDM symbols

resourceGrid = zeros(Nsym, Nsc);

% Downlink (DL) assignment: OFDM symbols 1 to 10
resourceGrid(1:10, :) = 1;

% Uplink (UL) assignment: OFDM symbols 11 to 14
resourceGrid(11:14, :) = 2;

% Plot the resource grid
figure;

imagesc(resourceGrid);

grid on;

xlabel('Subcarrier Index');
ylabel('OFDM Symbol Index');
title('Simplified 5G NR Time-Frequency Resource Grid');

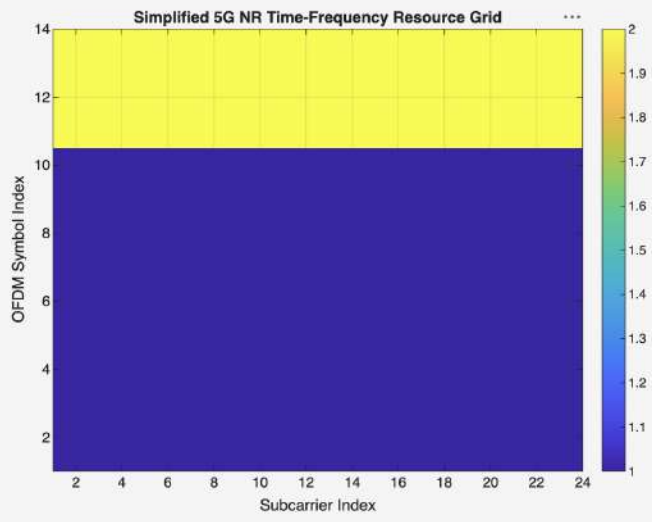
colorbar;

% Set axis limits
xlim([1 Nsc]);
ylim([1 Nsym]);

% Put OFDM symbol 1 at the bottom
set(gca, 'YDir', 'normal');
```

Figures

Figure 1



Command Window

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exp5.m | exp5b.m | exp6a.m | exp6b.m | exp7a.m | exp7b.m

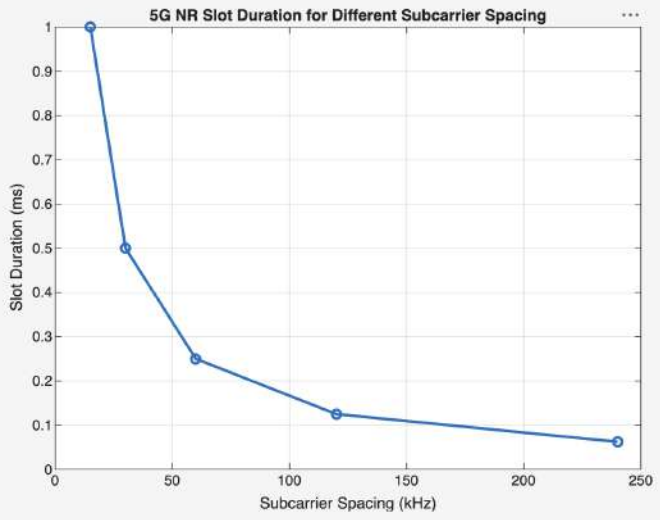
/MATLAB Drive/exp7b.m

```
clc; clear; close all;
scs = [15 30 60 120 240]; % kHz
mu = 0:4;
slot = [1 0.5 0.25 0.125 0.0625]; % ms

figure
plot(scs,slot,'o-','LineWidth',2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')
figure
bar(scs)
grid on
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```

Figures

Figure 1 | Figure 2



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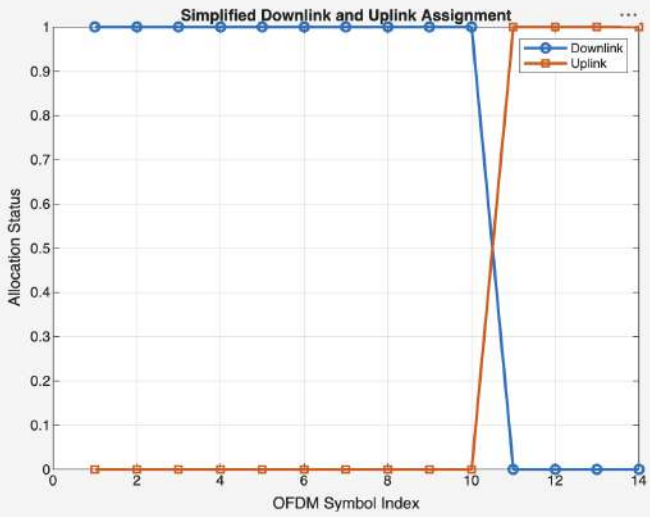
exp5.m | exp5b.m | exp6a.m | exp6b.m | exp7a.m | exp7b.m | exp7c.m

/MATLAB Drive/exp7c.m

```
clc; clear; close all;
symbols = 1:14;
DL = [1 1 1 1 1 1 1 1 1 1 0 0 0 0];
UL = [0 0 0 0 0 0 0 0 0 0 1 1 1 1];
figure
plot(symbols,DL,'o-','LineWidth',2); hold on
plot(symbols,UL,'s-','LineWidth',2)
grid on
xlabel('OFDM Symbol Index')
ylabel('Allocation Status')
title('Simplified Downlink and Uplink Assignment')
legend('Downlink','Uplink')
```

Figures

Figure 1



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